The Science: Gluten, Heat and Survival

What acid, enzymes and heat do to a dough and a culture

Sourdough Science

github.com/zrrbite/sourdough-science

What this deck covers

Seven practical questions, each answered with its mechanism and its source:

1. Why does sourdough make dough slack and open, and when does it turn fragile?
2. Do you need to knead and fold, or will time and the fridge do the work?
3. Can heat buy flavour, and where does it start to cost leavening?
4. How can a pure sourdough rise after hours at a 40°C+ setpoint?
5. How hot is the dough really, when the oven says 45°C?
6. How long can a culture take the heat, and how do you rescue one that didn't?
7. Is a hot-kept culture special, and which test would settle it?

Flags: **own result** the author's own kitchen results, kept apart from the literature; **hypothesis**, **contested**, **not yet tested** as in the notes. Temperatures are oven setpoints, not dough temperatures, unless stated.

Source: chapters 05–11 and two notes: *Why the Hot Protocol Works*, *The Levain Doubling Test*

Gluten

Gluten is springs plus lubricant

Glutenin: the springs

- Very large polymers: subunits linked end to end by S–S (disulfide) bonds
- Gives **elasticity**: resists stretching and snaps back

Gliadin: the lubricant

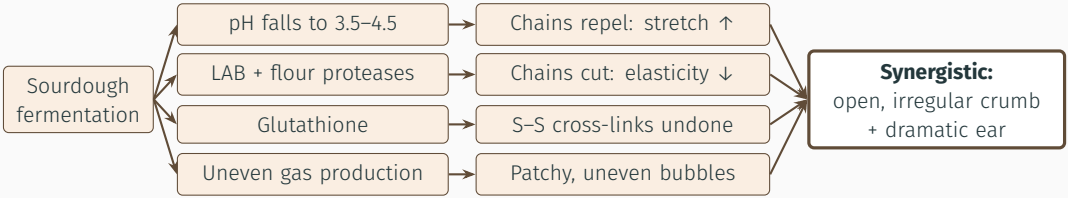
- Smaller single chains, folded by internal S–S bonds only
- Gives **extensibility**: lets the network flow and stretch

- More glutenin: strong, snap-back dough (high-protein bread flour). Less: softer, more extensible (Tipo 00, pastry flour)
- S–S bonds keep breaking and re-forming: why dough can be reshaped, and attacked

In the kitchen: acid, proteases and hydration changes all shift this balance. What you feel as stretch versus snap-back is this ratio moving.

Source: ch. 05; Wieser (2007); Shewry et al. (2002)

Fermentation loosens gluten four ways at once



- Acid scales with pH: modest at pH 5.0 (10–15% levain hybrid), dramatic at 3.8
- Proteases keep cutting, and flour proteases work faster at pH 3.5–4.5 than at neutral: modest after a 4 h bulk, extensive after 12 h bulk + 48 h retard
- **contested** Several studies find added acid firms dough; slackening is the fermentation case

In the kitchen: levain % sets how slack the dough gets; fermentation time decides when extensible tips into fragile. For pizza, that line is thin.

Source: ch. 05; Clarke et al. (2004); Thiele, Grassl & Gänzle (2004); acidification note in ch. 25

Time builds gluten without kneading

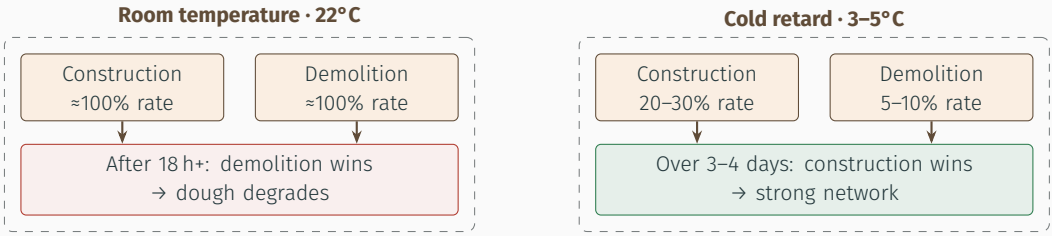
Mechanism	What it does	Time
Diffusion	Thiol groups meet and form S–S bonds, as in kneading, only slower	12–18 h at ≈22° C (no-knead); 3–5× slower at 3–5° C
Oxidation	Air folded in turns –SH into S–S all through the retard (what ascorbic acid and bromate speed up)	Builds over 3–4 days
Gas pressure	Each growing bubble stretches the gluten around it: a stretch-and-fold from inside, everywhere at once	Continuous, even in the fridge; possibly the largest contributor
Hydration	Water slowly unfolds the proteins, exposing buried bonding sites	More complete after 3–4 days

Kneading and folding only *accelerate* these reactions, and the slow network is arguably more uniform.

In the kitchen: one coil fold initialises the network: it evens out hydration, adds air for oxidation and gives an alignment bias. For a 72–96 h retard, one fold is enough.

Source: ch. 06

Cold slows demolition more than construction



- Building (diffusion, oxidation, gas pressure) has low activation energies and keeps going in the cold; demolition (proteases, glutathione) has higher ones, so the cold suppresses it disproportionately. The rates are rough estimates

In the kitchen: mix, fold once, ball and retard for 48–72 h. Hands-on time drops from 3–4 h to about 30 min, and the cold builds the network as it develops flavour.

Source: ch. 06

Heat as a flavour tool

Heat-stressed yeast make more flavour compounds, not fewer

- Usual ranges (levain 25–28°C, bulk 22–25°C) target schedule and leavening, not flavour
- Near their stress threshold (roughly 35–40°C for most strains) yeast mount a **heat shock response**: chaperone proteins (HSPs), trehalose, more saturated membranes
- Fusel alcohols rise (isoamyl alcohol, isobutanol, 2-phenylethanol): the precursors of fruity, floral esters. The ATF1/ATF2 ester enzymes also run faster
- LAB thrive where yeast are stressed (*L. delbrueckii* grows best at 42–45°C; many lactobacilli fully active at 35–40°C): more acid, proteolysis, LAB aromas
- Net: more flavour per hour, in a shorter window than 8–12 h at 25°C

In the kitchen: a warm levain is a flavour lever, not just a speed-up. The next slide is the catch.

Source: ch. 07; Verstrepen et al. (2003); Saerens et al. (2010); Kucharczyk & Tuszyński (2018); Landaud et al. (2001)

Hotter does not reliably mean more esters contested

More with heat: brewing

- Kucharczyk & Tuszyński (2018): esters and higher alcohols rose significantly with temperature, at industrial scale
- Landaud et al. (2001): temperature drives fusel alcohols independently of pressure

Fewer with heat: sourdough yeast

- Ferrández-Gómez et al. (2024): sourdough strains in beer wort at 37°C gave a *drastic reduction* in volatiles, esters especially

- Likely reason: heat speeds both ester synthesis and volatile loss. Which wins depends on strain, headspace, surface-to-volume and duration
- Brief, closed, dough-bound: likely a net gain. Long, open, runny: likely a loss

In the kitchen: don't assume hotter means fruitier: taste it against a room-temperature control. That side-by-side is still to be done. **not yet tested**

Source: ch. 07, caveat on conflicting evidence

The thermal risk gradient is a slope, not a cliff

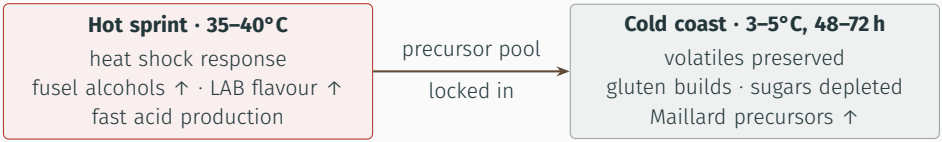
	Yeast	LAB	For the baker
25–30°C	Comfort zone; no stress response	Active	Most recipes
30–35°C	Near optimum; mild stress at the top	Accelerate in step	Faster; most cope
35–40°C	Full heat shock response; growth slows, metabolites rise	Thriving; may dominate	**The flavour sweet spot**
40–45°C	Most strains lose viability toward the top; a gradual decline	Thermophiles may stay fully active	Flavour; gas may fail
45°C+	Thermal kill for most yeast	Mesophiles decline; thermophiles may persist	Not useful for dough

Strain- and culture-dependent, from lab studies: a warm-kept or heat-selected culture may exceed these.

In the kitchen: if a 40–45°C build leavens sluggishly, the yeast have fallen below functional levels. The LAB still make flavour, so it tastes right but won't rise.

Source: ch. 07, from the literature on the most studied sourdough species

Hot sprint, cold coast: make flavour hot, keep it cold



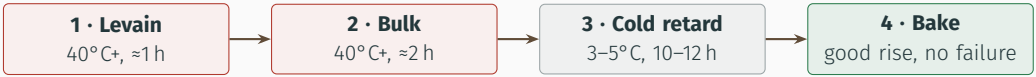
- The catch: more LAB activity means more proteases, so a hot front end loads gluten damage onto the dough before the fridge
- Limit it: keep the *dough* hot 30–60 min at most; use 10% levain rather than 15–20%; use a high-W flour (Caputo Oro W 350–380+). W 280–310 may not survive

In the kitchen: build the levain at 35–38°C: it peaks in 4–6 h rather than 8–12, and passes peak faster, so catch it right at peak. Mix the dough at ≈22°C, fold once, retard.

Source: ch. 07

The hot culture and Kveik

A pure sourdough rose after ≈3 h at a 40°C+ setpoint



own result Repeated with success: well-leavened crumb, no structural failure, and no commercial yeast. All the gas came from the culture's wild yeast.

- The literature predicts inhibition or death here: *K. humilis*, a textbook sourdough yeast, is inhibited at 35°C (Brandt et al. 2004)
- 40°C+ is the oven setting. The dough itself was never probed

In the kitchen: test your own culture rather than trusting species data. If a 40°C+ build gives a well-risen loaf, it is heat-tolerant enough for that protocol.

Source: ch. 08 (own kitchen results); Brandt et al. (2004)

Two explanations, and they fail in different ways

	1 · Genuine thermotolerance hypothesis	2 · Temporal escape hypothesis
The idea	The culture holds heat-adapted yeast, active and dividing at 40°C	Ordinary yeast, genuinely stressed; ≈3 h is short enough that enough survive
Gas comes from	Yeast still growing at temperature	Non-dividing cells still fermenting sugar, plus CO_2 dissolved in the cold
Why plausible	*Pichia kudriavzevii* survives 40°C in sourdough (Meroth et al. 2003); warm builds could select, as with Kveik	Heat kill is time-dependent: at 40°C, gradual rather than instant
Hot bulk stretched to 4–8 h	Still makes gas (gluten may fail first)	Leavening collapses abruptly, probably between 4 and 8 h

- not yet tested **Open.** Neither the extended-bulk test nor the levain doubling test has been run. A note bets on explanation 2 (last section)

Source: ch. 08; Meroth et al. (2003); Brandt et al. (2004)

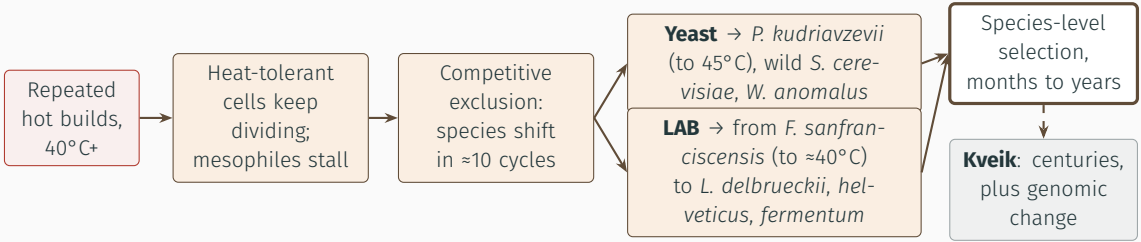
Warm practice selects microbes, over centuries or days

Study	What it found
KVEIK: farmhouse yeast pitched warm (often 28–40°C+) and reused from good batches	
Preiss et al. (2018)	Kveik from nine Norwegian sources: a genetically distinct group of domesticated *S. cerevisiae*, with marked ethanol and heat tolerance
Foster et al. (2022)	More trehalose than standard ale yeast: a direct heat-tolerance mechanism
SOURDOUGH: same backslopping, different temperatures	
Van der Meulen et al. (2011)	Wheat sourdough: 23°C selected *Leuconostoc citreum*; 30 and 37°C selected *L. fermentum*, a complete shift within 10 backslopping cycles
Urien et al. (2014)	Rye sourdough, 56 daily cycles: at 20°C *Kazachstania unispora* alone; at 30°C *S. cerevisiae*, *P. kudriavzevii* and *C. glabrata*

In the kitchen: your feeding temperature is a selection pressure: on its own it can replace a sourdough's dominant species in about ten feeds.

Source: ch. 09; Preiss et al. (2018); Foster et al. (2022); Van der Meulen et al. (2011); Urien et al. (2014)

At home, heat selects species, not new genes



- The LAB shift favours lactic over acetic: may suit pizza, as acetic tightens dough
- **hypothesis** That this has happened to the author's culture is ch. 08's open explanation 1

In the kitchen: selection only bites if you backslop from hot builds: 10–20 cycles reshape the culture. A mother kept at 20–25°C keeps its full consortium, so keep one as backup.

Source: ch. 08–09; Van der Meulen et al. (2011); Meroth et al. (2003); Limtong et al. (2018); Brandt et al. (2004)

Thermal lag

The oven setpoint is not the dough temperature

Dough's thermal diffusivity is like water's ($\approx 1.4 \times 10^{-7}$ m²/s); plastic insulates further.

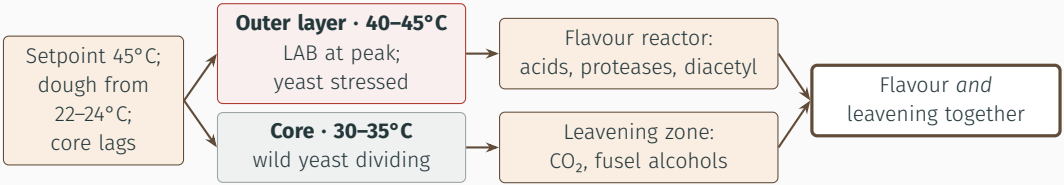
Vessel at a 45°C setpoint	How long the centre lags
Small jar, 75 g	Warms through in a small fraction of an hour
Levain jar, 200–300 g	May take 30–45 min to reach setpoint
Dough tub, 1 kg+	An hour or more

- "1 h at 40°C+", 200–300 g jar: core 25–35°C for the first half; average probably 30–38°C
- **not yet tested** At 40–50°C settings the core is likely 5–15°C below setpoint most of the time: an estimate, as no dough has been probed yet

In the kitchen: probe the centre of the dough and log it through a build: time to setpoint, time-weighted average, and the surface-to-core gap.

Source: ch. 10; Sadot et al. (2017); 75 g jar: *The Levain Doubling Test*; oven air measured ≈45°C on the usual setting (2026-09-13)

One dough, two fermentations: hot shell, cool core



- A uniform 40°C would cost leavening; a uniform 30°C would cost flavour
- The yeast may never feel what the dial implies, so heat selection is softer than the setpoint suggests (the shell is still above 40°C)

In the kitchen: if the unstirred hot build works, don't stir: it collapses the gradient, and its oxygen grows more yeast but fewer fusels and esters. To test, stir once at halfway.

Source: ch. 10

Thermal death and recovery

Thermal death is a rate, not a threshold

own result Brief excursions to 40°C+ (≈1 h levain, ≈2 h bulk, each then retarded cold) did no apparent harm over several days. One prolonged session at the same temperature killed, or very nearly killed, the culture.

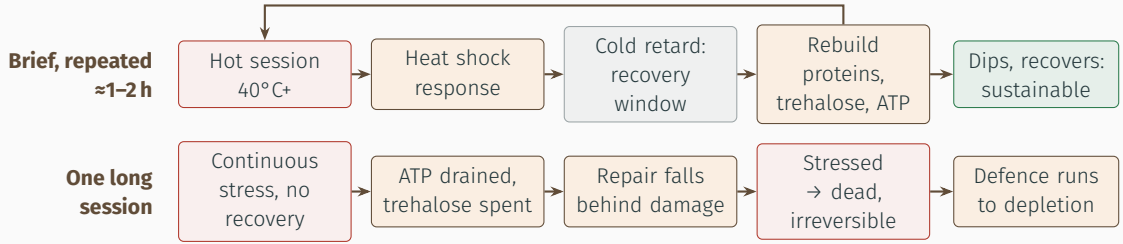
- Heat kills at a rate: numbers fall exponentially with time at temperature
- **D-value**: time for a 90% (one-log) kill. **z-value**: the rise that cuts D tenfold
- So 10 min at 45°C and 2 h at 42°C may kill about equally
- Lab D-values for *S. cerevisiae*: 22.3 down to 0.8 min across 45–55°C, in buffer or media. In dough they are lower bounds on survival: use qualitatively

In the kitchen: a temperature on its own says nothing about survival. Always give the time at temperature too.

Source: ch. 11; Guerrero, López-Malo & Alzamora (1999/2001)

The heat-shock defence is a budget, and the fridge refills it

- It works (a mild 37°C pre-shock induces tolerance to otherwise lethal heat; Morano et al. 2012) but burns ATP, uses up trehalose, and repairs damage without preventing it



In the kitchen: the cold retard after each hot session is part of the method, not a pause: it is where the culture recovers.

Source: ch. 11; Morano et al. (2012)

Two free protections: a slow ramp and a stiff levain

Ramp, not shock

- *S. cerevisiae* survives a later hour at 50°C considerably better after a slow, linear warm-up than after a sudden shock
- Gained mainly between 40 and 50°C on the way up, by membrane remodelling, not Hsp104 or trehalose
- So the container's lag protects: pre-warming it, or dropping a small levain into a hot water bath, is harsher than it looks

Lower water activity

- Heat resistance rises as water activity falls: a stiff levain does from outside what trehalose does inside
- 100% vs 50% hydration is perhaps a_w 0.99 vs 0.95: plausibly a factor of a few in D-value. **not yet tested** No published value at 40–45°C
- 100% to 85% buys very little; a true 50–60% stiff levain buys more

In the kitchen: start hot builds from cool, in their container, and stiffen the levain *before* raising the heat. A margin, not a licence: neither extends a safe session.

Source: ch. 11; Guyot, Ferret & Gervais; Morano et al. (2012)

Long sessions are harsher: acid, ethanol and hunger compound

- **Falling pH** speeds heat kill; pH control and heat defence share one ATP budget
- **Ethanol** speeds heat kill exponentially; both hit the membrane, so they multiply (Leão & van Uden 1982; Piper 1995)
- **Sugar runs out**: no fuel for the defence, no carbon to rebuild trehalose

"Yeast that comfortably survives 45°C at pH 5.5 with ample maltose may die at 45°C at pH 3.8 with none."

Regime	Assessment
Brief (1–3 h) at 40–45° C, recovery between	Sustainable, mildly selective: the documented working protocol
Extended (6 h+) at 40–45° C, no recovery	High risk: the defence exhausts, stressors pile up
Any duration at 50° C+	Thermal kill for most sourdough yeast
Hot session on an acidic, sugar-depleted culture	Substantially more lethal than on a fresh feed

In the kitchen: feed immediately *before* a hot session, not after, and build stiff. Neither extends the time a culture can take; both raise its odds.

Source: ch. 11; Leão & van Uden (1982); Piper (1995)

Sour but flat means the yeast died first

You see	It means
Sharp acid, no gas, no rise	**Yeast lost, LAB surviving.** The most common outcome
No acid, no gas, flat or “off” smell	Both badly damaged; poorer prognosis
Gas back within 1–2 feeds, acid intact	Reduced but viable; recovering
Sweet or solvent smell, no gas, slimy	Possible spoilage: discard

LAB are considerably more heat-tolerant, so a culture can smell healthy and sour with no leavening left.

Recovery: a 4–5 log kill leaves survivors

1. Feed at 22–25°C: warmth was the stressor
2. Feed ≥1:10:10 to dilute acid and ethanol
3. Add wholemeal or high-extraction flour
4. Feed daily for 3–7 days before judging
5. Pink, orange, fuzzy or putrid: discard

In the kitchen: keep the recovered culture as a second culture, not a replacement, and compare it with the original or a room-temperature backup.

Source: ch. 11

Why the hot protocol works

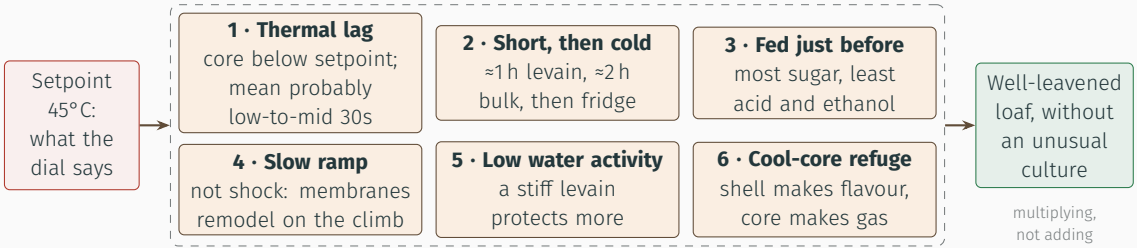
At 28°C nothing is stressed: the usual advice is about schedule

- *S. cerevisiae* grows best around 30–33°C; most sourdough LAB at 30–40°C. A levain at 28°C is near an **optimum**, not a limit
- Why the usual advice says 24–26°C:
  - **Schedule control**: cooler widens the timing window and forgives a late baker. The largest single reason
  - **A flavour preference stated as a rule**: warmer means more acid per unit of gas, and more lactic than acetic. A tradition chasing acetic sharpness runs cool
- It is not even universal: Italian *lievito madre* is classically kept around 28°C, and panettone practice runs at 26–30°C

In the kitchen: at 28°C you trade a narrower timing window and a rounder, more lactic sour. Not viability.

Source: *Why the Hot Protocol Works*, part 1; ch. 07

Six ordinary effects compound: no exotic culture needed hypothesis



- **The position**: temporal escape plus five other effects does nearly all the work. “It works” is weak evidence: a 90% kill still leaves enough to leaven

A stated bet, not a result. Heat-tolerant selection is at most a small extra term; the test that would falsify the bet has not been run.

Source: *Why the Hot Protocol Works*, parts 2–3; ch. 08, 10, 11

A doubling levain shows gas, not cell division not yet tested

One feed split into two equal jars: one hot, one at room temperature.

Hot jar does	It supports
Strong, prompt rise, like the control	Yeast active at temperature: **consistent with either explanation**
Acid and aroma, little or no rise	LAB active, yeast not contributing gas: temporal escape
Rise markedly slower than the control	Yeast impaired but working: temporal escape

Stressed, non-dividing yeast still make CO_2. A negative result is informative; a positive one is **not decisive**.

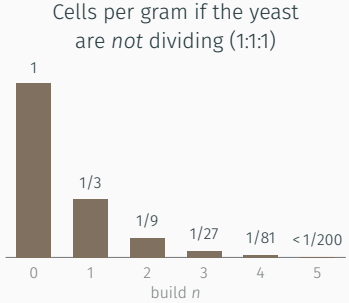
At the bench

- Same mother, same feed, **equal mass**: mass drives the thermal lag
- Identical tall, narrow jars; rubber band at the start line
- Control at room temperature, *measured* and written down
- Height every 15 min (photo with a ruler); oven air each time; core probe if you have one
- Feed the mother the day before; run until the hot jar peaks and recedes

Source: *The Levain Doubling Test*; ch. 08

Serial hot passage is the test that would settle it not yet tested

- At a 1:1:1 feed the old starter is a third of the new mass
- If the yeast are not dividing, cells per gram fall to $(1/3)^n$: a twenty-seventh by build 3, under a two-hundredth by build 5
- Five hot builds in a row, each backslopped **from the hot jar**, not the fridge or the mother; same ratio, time, setpoint
- If build 5 still rises like build 1, the yeast are dividing at temperature: explanation 1, and the bet is wrong



In the kitchen: split off a test jar first. Never run this on the only culture in the house.

Source: *The Levain Doubling Test*; ch. 08

Takeaways

- Acid, proteases and glutathione slacken gluten: levain % sets how much, fermentation time decides when extensible turns fragile
- Time and cold build gluten: one fold and a 48–72 h retard replace most kneading
- Heat can buy flavour (35–40° C), but more esters is *not* guaranteed: taste against a control, and keep the dough's own hot phase short
- The dial is not the dough: a 200–300 g jar's centre may lag 30–45 min, a 1 kg tub an hour or more
- Survival is time at temperature: brief hot sessions with a cold recovery are sustainable, long ones high risk
- Feed right before the heat, start from cool, build stiff, keep a room-temperature backup
- Sour but flat means the yeast died: recover at 22–25° C, 1:10:10, for 3–7 days

Open questions

Question	Test	Status
Heat-adapted, or temporal escape? (ch. 08; a note bets on escape)	Serial hot passage (decisive); levain doubling; extend the hot bulk until leavening fails	not yet tested
How hot is the dough really? (ch. 10)	Log the jar core in a hot build; 75 g jar vs 1 kg tub	not yet tested
Does heat raise or lower esters? (ch. 07)	Taste side by side with a room-temperature control	contested
How much does a stiff levain protect? (ch. 11)	Stiff vs liquid, same setpoint and time; then feed and time to peak	not yet tested
Is the recovered culture better? (ch. 11)	Against a control: heat tolerance, flavour, timing at 25°C	not yet tested
Stir the hot build or not? (ch. 10)	One stir at halfway vs none: rise and flavour	suggested

Source: ch. 07, 08, 10, 11; notes; README status list. The `chamber/` logger meant to run these has its first stage built; nothing is instrumented yet

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Full list: `docs/25-references.md`